MTH 200 Chapter 4: Applications of Derivatives

4.5 Worksheet

1. Find the intervals on which $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is increasing, decreasing, and local extrema.

2. Suppose the function *f* has the first derivative of $f'(x) = e^{2x}(x^2 + 25)(x-3)^2(x^2 - 64)(2x-1)$.
   List each interval upon which the function *f* is increasing and decreasing. State the value of *x* at which any local maximums or minimums of *f* occur.

3. Find all intervals on which $f(x) = 3x^5 - 10x^4 + 10x^3 - 40x + 50$ is concave up and down. Find all inflection points on the graph of *f*.

4. Find the critical numbers of $f(x) = (x-1)^2\sqrt{x}$.

5. Suppose that $g(x)$ is a function whose first derivative is $g'(x)=\dfrac{8e^{x}\left(x^{2}+36\right)\left(x^{2}-25\right)(x-4)^{10}}{\ln\left(x^{2}+9\right)}$.

Find all critical numbers of $g(x)$. At each critical number, state whether $g(x)$ has a local maximum, local minimum or neither at that point.

6. A polynomial $f(x)$ has the second derivative $f''(x)=xe^{x}(x+7)^{6}\left(x^{2}-196\right)\left(x^{2}+25\right)$. State the interval where *f* is concave up and concave down. State each *x*-value at which the graph of $f(x)$ has an inflection point.

7. Find each interval of concavity and each inflection point for the function $f(x)=\frac{x-5}{x^3}$.

8. Find when the function $f(x)=8+5xe^{-3x}$ is increasing, decreasing, concave up, and concave down. Also find all relative extrema and inflection points of $f(x)$.

9. A bullet is fired upwards from the ground at an initial velocity of 1200 feet per second. The height s (in feet) of the bullet above the ground after t seconds is $s = 1200t - 6t^2$ on Mars and $s = 1200t - 16t^2$ on Earth. How much higher will the bullet travel on Mars than on Earth? (Hint: What is the slope of the tangent line when the bullet is at it highest point?)

10. A rock is thrown upward from the surface of the moon. Between the time that the rock is thrown and the time that the rock hits the ground, the rock's height is given by the formula $s(t) = 24t - 0.8t^2$, where t is the number of seconds since the rock is first thrown and $s(t)$ is measured in meters above the moon's surface.

a. Find a formula for the rock's velocity at time t.

b. Find a formula for the rock's acceleration at time t.

c. How fast was the rock moving, and in which direction, at the end of 25 seconds.

d. What is rock's maximum height and at what time does it reach that height?

11. Sketch a graph of the curve $f(x) = x^5 - 15x^3$.

12. Find the absolute extrema of $f(x)=\frac{1}{x^2-x}$ on the interval $(0,1)$.

13. Find the absolute extrema of $f(x)=e^{\left(x^3-3x^2\right)}$ on the interval $(0,\infty)$.

14. Determine the absolute minimum y-value on the graph of $y = 2e^{4x} - 648x + 38$. Simplify your answer fully.

15. Determine the x-value at which the function $f(x) = \dfrac{e^x}{e^{2x} + 5}$ attains its absolute maximum value.

16. What are the coordinates (x, y) for the highest point on the graph of the function $g(x) = 10xe^{-2x}$.

17. Find the intervals where $f(x) = xe^{x}$ is increasing, decreasing, concave up, and concave down. Find all local extrema and inflection points of *f*.

18. Find the intervals where $f(x) = x + 2\sin(x)$ is increasing, decreasing, concave up and concave down on the interval $[0, 2\pi]$. Find all the local extrema and inflection points of f.